

Gearing Decides the Sign: An Adaptive, Self-Tuning Parallel Knee Preload for Energy-Efficient Quadruped Load Carriage

I. Osokin *et al.*

Abstract—Adding a passive parallel elastic element beside a motor offloads joint torque and, because resistive (ohmic) motor loss scales with the *square* of torque, should cut electrical energy. Whether it does, however, is governed by the actuator’s gear ratio, which sets how large a share of the electrical budget the ohmic term occupies. We study this trade-off end-to-end in simulation (MuJoCo Playground / MJX, brax PPO) on two commercial Unitree platforms. On the high-gear G1 humanoid (ohmic $\approx 4\%$ of the budget) an in-loop parallel hip spring makes walking *worse* ($+7\%$), reversing an optimistic post-hoc estimate. On the low-gear Go1 quadruped (ohmic $\approx 54\%$) a *constant* parallel knee preload cuts the cost of transport (CoT) by 14–27% across three of four training conditions, with the benefit growing with payload. We further introduce an *adaptive, per-leg, self-tuning* preload: a slow outer loop that senses each knee’s own motor torque and ramps a passive preload to offload it, using no load sensor and no payload observation; it carries the energy win into blind load-carrying (0–6 kg). We report the boundary where the mechanism reverses, the constant-preload (vs. linear-spring) element insight for the offset-dominated knee, and the high-load stability cost honestly. The result is a deployable, sensor-free efficiency mechanism for low-gear load-carrying quadrupeds, and a clear statement of when parallel elasticity helps and when it hurts.

Index Terms—parallel elastic actuator, cost of transport, legged robots, quadruped, reinforcement learning, energy efficiency.

I. INTRODUCTION

Electric legged robots spend much of their energy as heat in the motor windings. For a geared joint the copper loss is $P_\Omega = (\tau/K_t)^2 R$, quadratic in joint torque τ , so any mechanism that offloads torque from the motor cuts heat super-linearly. A *parallel* elastic element—a spring beside the motor—does exactly this for the quasi-static support torque a leg holds during stance, and additionally stores and returns energy the motor would otherwise dissipate (most commercial drivers do not regenerate).

The catch is twofold. First, a parallel spring is *always engaged*: it also resists the motor in phases where its torque is unwanted, so the policy must learn to exploit it. Second, and decisively, the ohmic term is only worth attacking if it is a large share of the budget—and that share is set by the **gear ratio**. High reduction makes the motor-side torque small and the ohmic loss negligible; low reduction leaves large torque at the motor, where offloading pays.

Contributions. (1) We show *gearing decides the sign*: the same parallel-spring idea is harmful on a high-gear humanoid

and beneficial on a low-gear quadruped, quantified in matched in-loop experiments. (2) For the low-gear Go1 quadruped we measure a **14–27% cost-of-transport reduction** from a *constant* parallel knee preload, and show the element must be a constant preload, not a linear spring—the knee work-loop is offset-dominated. (3) We introduce an **adaptive, per-leg, self-tuning preload** that requires no load sensor and no payload observation, and carries the energy win into blind load-carrying with the benefit *growing with load*. (4) We report the limits honestly: a seed-dependent magnitude, a high-load stability cost, and a sim-to-real realism bound on the payload range.

II. RELATED WORK

Parallel-elastic torque offloading to cut motor energy is well established, and the literature reveals a pattern that independently corroborates our thesis: *the large wins are all on low-gear, backdrivable, ohmic-dominated systems*. A passive clutched ankle spring in parallel with the calf cut human walking metabolic cost by 7.2% with zero actuation [1]; Sandia’s STEPPR placed parallel springs at the hip and ankle for a predicted 30–50% reduction in cost of transport [2]; and on the ANYmal quadruped a knee parallel spring co-optimized with an RL controller improved torque-squared efficiency by 33% and cut peak joint torque by 30% [3]—the closest prior art to ours. Clutched/quasi-passive parallel actuators (the CPEA [5]; BirdBot’s self-engaging clutch reducing required knee torque $\sim 10\times$ [4]) show that gating the spring to the load phase recovers most of the benefit; an *always-engaged* spring fighting the swing phase is precisely the failure mode we observe on the high-gear G1. For running, energy-aware MPC with parallel springs reports 38.8% CoT in simulation but only 14.8% on hardware [6]—a sobering sim-to-hardware ratio we keep in view. Tunable parallel springs optimized with a torque-squared metric were studied analytically on a leg test-stand by the present group [7], without RL, gait, or load adaptation; RL with elastic actuators has otherwise been explored mainly on quadrupeds [8].

Our contribution is therefore one of *integration and application*, not a new mechanism: a commercial low-gear platform, in-loop RL gait co-adaptation, electrical/ohmic cost-of-transport accounting, the explicit post-hoc-vs-in-loop and G1-vs-Go1 gearing contrast, and—the distinguishing element—a sensor-free, per-leg, *load-adaptive* preload. We deliberately do *not* add a clutch (which the literature identifies as the natural

way to defeat the swing-phase penalty); our adaptive preload instead tracks only the slow support component the always-on spring can safely carry, which is what makes it sensor-free and deployable. A fuller survey—series vs. parallel elasticity, variable-stiffness actuators, clutched/quasi-passive elements, and biological running springs—is in the supplementary literature review (docs/literature_review.md).

III. METHOD

A. Platforms, simulator, and policy

We use MuJoCo Playground’s Unitree Go1 (12 kg, 6.33:1 reduction) and G1 (high reduction) joystick environments, simulated with MJX and trained with brax PPO (8192 parallel environments, 200 M steps). The policy is a feed-forward actor-critic tracking a commanded body velocity at 50 Hz, under standard domain randomization.

B. Parallel spring model and honest energy accounting

The spring torque $\tau_{\text{spring}}(\theta)$ is injected at the target joint through the external generalized force `qfrc_applied`, *not* through the actuators, so the motor torque `qfrc_actuator` excludes the passive contribution and the electrical accounting stays honest. Per actuated degree of freedom the electrical power is

$$P_{\text{elec}} = \max(\tau \omega + (\tau/K_t)^2 R, 0), \quad (1)$$

i.e. mechanical plus ohmic with *no regeneration* (the $\max(\cdot, 0)$ clamp), applied identically in the training reward and the evaluation metric. The headline metric is cost of transport, $\text{CoT} = P_{\text{elec}}/v$ (electrical watts per metre per second), which normalizes for the speed differences the spring induces.

C. The element: a constant preload, not a linear spring

Fitting a linear spring to the Go1 knee work-loop degenerates to $k \approx 0$: the loop is dominated by a near-constant support offset through stance, not a restoring slope (a τ^2 fit is even mildly anti-restoring). The buildable optimum is therefore a *constant preload*, realizable as a heavily pre-wound low-rate coil whose torque is nearly constant over the knee’s small operating range.

D. Adaptive, per-leg, self-tuning preload

A slow outer loop wraps the fast policy. For each leg i it maintains a 15 s exponential moving average $\bar{\tau}_i$ of that knee’s *own* motor torque and ramps the passive preload toward full compensation,

$$\dot{\tau}_{0,i} = \text{clip}(k_p \bar{\tau}_i, -2, +2) \text{ N m/s}, \quad k_p \approx 0.2, \quad (2)$$

with no load sensor and no payload observation. The four legs are independent reflex loops, so front/rear and left/right asymmetry is handled automatically. Because training episodes are shorter than the 15 s window, we *train robust* (preload domain-randomized per episode) and *adapt at deploy* (run the controller in long rollouts so τ_0 converges); the converged τ_0 scales with load and is larger on the front legs.

TABLE I
COST-OF-TRANSPORT REDUCTION (%), ADAPTIVE KNEE PRELOAD VS. MATCHED NO-SPRING BASELINE, FLAT GROUND.

Condition	@0 kg	@2.5 kg	@5 kg
Seed 1	−16.6	−19.5	−22.8
Seed 2 (weak outlier)	−3.4	−8.3	−6.5
Seed 3	−13.9	−20.1	−26.7
Payload curriculum	−16.8	−20.4	−22.0

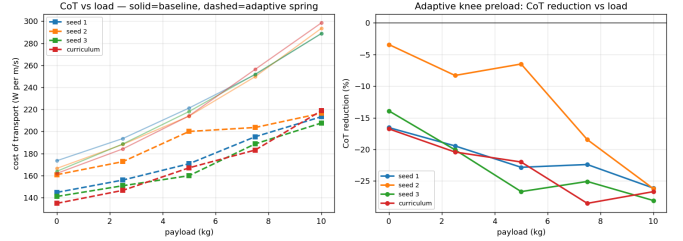


Fig. 1. Cost of transport (left; solid = baseline, dashed = adaptive) and its reduction (right) vs. payload. The benefit grows with load in three of four conditions; seed 2 is a weak outlier.

IV. EXPERIMENTS AND RESULTS

A. Gearing decides the sign

On the high-gear G1, where ohmic loss is $\approx 4\%$ of the motor budget, an in-loop parallel hip spring makes walking **+7% worse** and less stable, reversing an optimistic post-hoc -3.8% (the always-on spring fights the gait with no benefit to recover). On the low-gear Go1, ohmic is $\approx 54\%$ of the budget and the knee preload pays (below). This sign flip with gear ratio is the central organizing result.

B. Go1 load-carrying: cost of transport vs. load

Table I reports the CoT reduction of the adaptive preload against a matched no-spring baseline on flat ground, for three seeds and a payload curriculum; Fig. 1 plots CoT and the reduction vs. load.

The adaptive preload cuts CoT by **14–27% in three of four conditions**, and the reduction *grows with payload*—heavier loads raise the support torque the preload offloads. One seed (seed 2) is a weak -3 to -8% outlier: with the offload available, that policy spends it on walking faster rather than on energy, which the speed-normalized CoT only partly compensates.

V. DISCUSSION AND LIMITATIONS

We foreground the threats to validity rather than bury them. **(1) Stability cost.** At high load (≥ 7.5 kg) the adaptive policy loses some survival (e.g. 1070/1500 at 10 kg) where the matched baseline holds 1500/1500; the energy win is a low-to-mid-load result. **(2) Realism bound.** The real Go1 carries ~ 5 – 10 kg; the simulator enforces peak but not continuous/thermal torque, structure, or balance, so high-load (15 – 30 kg) “walking” is sim-only. The physically meaningful range is 0 – 6 kg; the 15 – 30 kg regime would belong to a B2-class robot. **(3) No-regeneration dependence.** The win lives partly in dissipated braking energy; with regeneration it shrinks

($\sim 24\%$ sensitivity). No-regen is justified by back-EMF physics for these drivers but is not a measured spec. **(4) Absolute energy is a band.** We lack measured motor K_t, R ; relative CoT comparisons are robust, but absolute watts are uncertain. **(5) Energy-naive baseline and rough terrain.** The baseline does not itself optimize energy hard, and on rough (2.5 cm) terrain both arms are unstable ($\sim 40\%$ survival), so the win there is inconclusive.

VI. CONCLUSION

Whether a parallel elastic element saves electrical energy in a legged robot is decided by the gear ratio: it is harmful on a high-gear humanoid and beneficial on a low-gear quadruped. For the low-gear Go1 we demonstrate a 14–27% cost-of-transport reduction from a constant knee preload, and a sensor-free, per-leg, self-tuning adaptive preload that carries the win into blind load-carrying with the benefit growing with load. The mechanism is simple and deployable; its boundary, its element form, and its high-load stability cost are characterized. Future work runs the same idea on a *running* quadruped, where impact braking energy is largest and a one-sided stiffness spring may add a recovery channel a constant preload cannot. The literature’s standard remedy for the always-on penalty—a clutch that disengages the spring during swing—is the clearest path to extend the win into the high-load regime where stability currently degrades, at the cost of the sensor-free simplicity our adaptive preload preserves.

REFERENCES

- [1] S. H. Collins, M. B. Wiggin, and G. S. Sawicki, “Reducing the energy cost of human walking using an unpowered exoskeleton,” *Nature*, vol. 522, pp. 212–215, 2015.
- [2] A. Mazumdar *et al.*, “Parallel elastic elements improve energy efficiency on the STEPPR bipedal walking robot,” *IEEE/ASME Trans. Mechatronics*, 2017.
- [3] F. Bjelonic *et al.*, “Learning-based design and control for quadrupedal robots with parallel-elastic actuators,” *IEEE RA-L*, 2023, arXiv:2301.03509.
- [4] A. Badri-Spröwitz *et al.*, “BirdBot achieves energy-efficient gait with minimal control using a self-engaging mechanical clutch,” *Science Robotics*, vol. 7, no. 64, 2022.
- [5] M. Plooi and M. Wisse, “A novel spring mechanism to reduce energy consumption of robotic arms,” *IEEE/RSJ IROS*, 2012. (The frequently-cited $\sim 80\%$ figure is disputed; an adversarial re-check traces the 2012 paper to $\sim 20\%$ on a 2-DOF arm.)
- [6] “Energy-aware kinodynamic MPC for legged robots with parallel elasticity,” arXiv:2503.05666, 2025 (38.8% CoT in sim, 14.8% on hardware).
- [7] A. Belov, I. Osokin *et al.*, “Torque-squared-optimal parallel elastic elements for a robotic leg,” Skoltech, 2024.
- [8] F. Bjelonic *et al.*, “Reinforcement learning for elastic quadrupedal locomotion,” 2022.